EmbodiedX Real-Time Embodied AI Inference Benchmark

Methodology Specification and Literature Synthesis Across Single-Chip Cloud GPU and TPU Accelerators

CoRL 2026 Workshop on Efficient Foundation Models for Real-Time Embodied AI (EEAI)

Organizing Committee Specification Report (v1.0.0) · `efficient-embodied-ai/embodied-efficiency-bench` · https://efficient-embodied-ai.github.io/challenge/

1. Motivation and Season 1 Design Summary

Embodied AI foundation models span Vision-Language-Action (VLA) control policies (π_0 [6], $\pi_{0.5}$ [7], `GR00T N1 / N1.7-3B` [8], `SmolVLA` [24]) and action-conditioned world models (`Cosmos-Predict2.5-2B` [25], `Genie Envisioner` [27]). Recent inference work accelerates these models through custom device kernels (`Pallas`, `Triton`, `CUDA`), cross-frame visual token caching (`VLA-Cache` [42]), numerical quantization and compression (`FP8`, `INT8`, `INT4`, `EfficientVLA` [43, 52, 53]), and asynchronous action-chunk scheduling (`RTC` [10], `TT-RTC` [18], `VLASH` [17]).

Standard robotics leaderboards evaluate models in stop-and-wait simulators where physics pauses during neural inference. Under a frozen simulation clock, a 2,000 ms policy and a 20 ms policy experience identical zero actuation delay, which hides physical control failures caused by object slip or contact drift mid-trajectory. Meanwhile, modern ML serving benchmarks such as **SemiAnalysis InferenceX (InferenceMAX)** [1, 2] and **MLPerf Inference** [35] plot **Throughput (`tok/s/chip`) vs. Interactivity (`tok/s/user`)** Pareto frontiers for autoregressive token streaming, whose axes do not capture receding-horizon action chunking or closed-loop actuation deadlines.

EmbodiedX (`embodied-efficiency-bench` v1.0.0) defines an open-source closed-loop serving benchmark for the **CoRL 2026 EEAI Workshop**. Season 1 executes every submission on one physical chip (`num_devices == 1`, TP = PP = EP = 1) across two standardized cloud platforms: **NVIDIA L40S (48 GB GDDR6)** and **Google Cloud TPU v6e-1 (32 GB HBM2e)**. Every evaluation trace is drawn deterministically from public real-world datasets (`HuggingFaceVLA/libero` and `lerobot/droid_100`), with zero synthetic sine-wave traces permitted in submissions. Under Option A launch neutrality, Season 1 opens on October 30, 2026 with 0 organizer-seeded vendor benchmark entries on the leaderboard so community pull requests establish the founding Pareto frontier directly.

1.1 Ten Architectural Decisions for EmbodiedX v1.0.0

EmbodiedX enforces ten architectural decisions (**D1**–**D10**, recorded in `docs/DESIGN_DECISIONS.md`):

1. **D1 (Closed and Open Divisions Under Locked K_{eval} and Host-CPU Offload Ban).** Both `Closed` and `Open` divisions lock execution horizon K_{eval} (5 actions on Track 1A, 8 actions on Track 1B, 16 predicted frames per rollout prediction on Track 2) so aggregate output cannot be inflated by lengthening open-loop horizons. Across both divisions, 100% of neural forward-pass operators (`MatMul`, `Attention`, `Conv`, `MLP`, `VAE`, text encoders) inside timed `act()` and `rollout()` calls must execute on the single target accelerator; host CPU threads handle only I/O, tokenization, RPC dispatch, and action queuing, while constant episode prompt embeddings may be pre-cached on the target accelerator during untimed `reset()`. **Closed Division** keeps checkpoint architecture and pretrained weights fixed at reference solver step counts under BF16/FP16/FP8 numerics checked by inline Golden-IO parity (≥ 100 real trace samples with mean quality floors and CVaR-1% worst-tail floors) and forbids pseudo-causal sub-clip slicing (`stream_chunk_frames > 0`) on Track 2. **Open Division** permits INT4 quantization, pruning, step distillation, or token merging when paired with a signed Stage-2 quality receipt.
2. **D2 (Three Independent Track Pools Anchored to Gapless Real-Time SLAs and Pinned Checkpoints).** At a 20 Hz servo loop ($\Delta t = 50$ ms, where 20.0 Chunks/s is the single-tick reaction ceiling), executing K_{eval} steps covers $K_{\text{eval}}\Delta t$:
   - **Track 1A (`pi0.5-LIBERO`).** Physical Intelligence $\pi_{0.5}$ [7] loaded from canonical ungated checkpoint `lerobot/pi05_libero_finetuned_v044` (commit `7152f13e34c6c115d808aea0e5d08aca74eac14b`, with pre-fit normalizers; 10 Euler ODE steps, dual 224×224 RGB traces from `HuggingFaceVLA/libero` commit `86958911c0f959db2bbbdb107eb3e17c5f9c798e`, $H = 10$, locked $K_{\text{eval}} = 5$, ≈ 0.734 TF/action), gapless SLA $L_{p99} \leq 250.0$ **ms** (≥ 4.0 **Chunks/s**). Both `PyTorch` (`CUDA` / `TorchTPU`) and `openpi JAX` are certified reference runtimes.
   - **Track 1B (`GR00T N1.7-3B`).** NVIDIA `GR00T N1.7-3B` [8] loaded from `nvidia/GR00T-N1.7-LIBERO` (`libero_10` sub-folder, commit `2ea293aa20ba7cf5bbf3ba17a5fbcb1a01cbfe21`; `Cosmos-Reason2-2B` VLM + 16-step DiT flow action head, dual 256×256 RGB traces from `HuggingFaceVLA/libero`, $H = 16$, locked $K_{\text{eval}} = 8$, ≈ 0.404 TF/action), gapless SLA $L_{p99} \leq 400.0$ **ms** (≥ 2.5 **Chunks/s**). Track 1A and Track 1B maintain separate award pools because `GR00T N1.7-3B` executes 1.82× fewer FLOPs per action than $\pi_{0.5}$.
   - **Track 2 (`Cosmos-Predict2.5-2B`).** NVIDIA `Cosmos-Predict2.5-2B` [25] loaded from checkpoint `nvidia/Cosmos-Predict2.5-2B` (`robot/action-cond` action-conditioned weights, commit `85f8ae7bfe8f5525c8d103429524dcf12f98bf7b`; bidirectional 3D-DiT video world model, 10 steps, CFG = 1.0, real-world 256×256 RGB manipulation traces from `lerobot/droid_100` commit `87301a2d2e99340e2010c9ef0f1d8e780b08aaf9`, locked $K_{\text{eval}} = 16$ frames per rollout prediction), gapless SLA $L_{p99,\mathbf{WM}} \leq 1000.0$ **ms** (≥ 1.0 **Pred/s**).

3. **D3 (Quality Verification as a Hard Admissibility Gate).** Golden parity sets require `--num-samples >= 100` real trace frames. In `Closed`, Track 1 enforces inline Golden-IO CosSim ≥ 0.995 plus CVaR-1% floor, and Track 2 enforces SSIM ≥ 0.88, PSNR ≥ 26.0 dB, CVaR-1%(SSIM) ≥ 0.80, and CVaR-1%(PSNR) ≥ 22.0 dB. In `Open`, Track 1 requires a signed Stage-2 receipt establishing latency-coupled `LIBERO-RT` success within -5.0 pp of the BF16 reference under actuation delay $d = \lceil L/\Delta t \rceil$; Track 2 requires SSIM ≥ 0.88, PSNR ≥ 26.0 dB, CVaR-1%(SSIM) ≥ 0.80, CVaR-1%(PSNR) ≥ 22.0 dB, LPIPS ≤ 0.12, FVD ≤ 120, inverse-dynamics consistency InvDyn ≥ 0.99, and policy-ranking correlation Kendall $\tau \geq 0.80$.
4. **D4 (Ordered Discrete Concurrency Grid, Pooled Tail Latency, and Stock-TDP Discipline).** LoadGen sweeps canonical concurrencies $C \in \{1, 2, 4, 8, 16\}$ starting at $C = 1$ (permitting only monotonic tail truncation on device OOM, with zero off-grid concurrencies and zero curve interpolation). Each point runs $N = 3$ trials ($N \geq 320$ chunks/run on Track 1, $N \geq 200$ predictions/run on Track 2, with untimed keep-alive drain) measuring pooled steady-state tail latency $L_{p99}(C)$ across all active slots paired with a mandatory $\leq 15\%$ per-slot completion fairness fence, within $\leq 8\%$ inter-run $p99$ spread ($\leq 5\%$ on $p50$/throughput) and $p99/p50 \leq 1.5$ at $C = 1$. Entrants run `--warmup >= 30` calls on the real model workload at stock vendor boost clocks and stock board TDP first; SM clocks are locked via `nvidia-smi -lgc` (calibrated directly on the live model loop instead of synthetic GEMM burners) only when dynamic boost trips the variance gate. Above-stock TDP or clock overrides are banned.
5. **D5 (Disambiguated Interactivity and Throughput Axes).** X plots Guaranteed Interactivity $X(C) = 1000/L_{p99}(C)$ in `Chunks/s` (Track 1) or `Predictions/s` (Track 2, with locked $K_{\text{eval}} = 16$ frames per rollout prediction). Y plots Fleet Throughput in steady-state wall-clock `Actions/s` (Track 1) or `Predictions/s` (Track 2) and hardware-measured efficiency in `Actions/J` or `Predictions/J` (`10 Hz NVML` constant-cap board power on `L40S`; `TPU v6e-1 = N/A`), plus auxiliary `$/hr` and Roofline `Iso-Compute` / `Iso-Bandwidth` views.
6. **D6 and D7 (Forward-Only Async Closed-Loop Timing and Control Diagnostics).** `LIBERO-RT` advances physics by $d = \lceil L/\Delta t \rceil$ ticks under a forward-only async loop without forcing +27% `IT-RTC vjp` backpropagation overhead [10, 18]. Every point co-reports Peak Age of Information $\Delta_{\text{peak}}(C, K_{\text{eval}}) = L_{p99}(C) + (K_{\text{eval}} - 1)\Delta t$ (ms), $p99/p50$, max $/p50$, per-slot completion skew ($\leq 15\%$), normalized action jerk, $\text{TTFF}_{p99} \equiv L_{\text{clip}}$ (Track 2), and peak device memory (GB).
7. **D8, D9, and D10 (Standardized Cloud Targets, Scalar Awards, and Commit-Reveal Governance).** Season 1 standardizes strictly on `NVIDIA L40S (48 GB)` (`nvidia_l40s_48gb`) and `Google Cloud TPU v6e-1 (32 GB)` (`tpu_v6e_1_32gb`), awards three scalar categories per track tie-broken by lower-step Real-Time Dominated Hypervolume wrt $(0.5X_{\text{SLA}}, 0)$, enforces per-frame cryptographic $\pm 1/255$ `uint8` pixel micro-jitter plus max $/p50 \leq 4.0$, and decides Final Official Season 1 Awards strictly via Phase-2 organizer clean-hardware re-runs on a private held-out split ($\pm 8\%$ Phase-1 check serves strictly as an anti-fraud disqualification gate).

2. Throughput versus Tail Interactivity in Closed-Loop Control

Serving LLMs trades aggregate **Throughput (`tok/s/chip`)** against per-user **Interactivity (`tok/s/user`)** [1, 2]. Serving embodied policies and world models obeys a corresponding tension between **Fleet Throughput (`Actions/s` or `Predictions/s`)** and **Guaranteed Interactivity (`1000 / p99_ms`)** across $C \in \{1, 2, 4, 8, 16\}$ (D5):

- **Guaranteed Interactivity (X-Axis, Right is Faster $\rightarrow$).** Captures pooled steady-state $p99$ completion rate across concurrent active streams $C \in \{1, 2, 4, 8, 16\}$ subject to the $\leq 15\%$ per-slot completion fairness fence:

$$X_{\text{VLA}}(C) = \frac{1000}{L_{p99}(C)} \text{ [Chunks/s]}, \qquad X_{\text{WM}}(C) = \frac{1000}{L_{p99,\text{WM}}(C)} \text{ [Predictions/s]} \tag{1}$$

  where $L_{p99}(C)$ is pooled steady-state $p99$ request latency in ms across $N \geq 320$ timed chunks (Track 1) or $N \geq 200$ timed rollout predictions of $K_{\text{eval}} = 16$ frames each (Track 2).
- **Fleet Throughput and Energy Efficiency (Y-Axis, Up is Higher $\uparrow$).** Captures steady-state full-occupancy wall-clock output under locked horizon K_{eval}:

$$Y_{\text{VLA}}(C) = \frac{N_{\text{timed}}(C) \cdot K_{\text{eval}}}{T_{\text{steady}}(C)} \text{ [Actions/s]}, \qquad Y_{\text{WM}}(C) = \frac{N_{\text{timed}}(C)}{T_{\text{steady}}(C)} \text{ [Predictions/s]} \tag{2}$$

  evaluated alongside measured energy efficiency $E(C) = Y(C)/\bar{P}_{\text{board}}(C)$ in `Actions/J` (Track 1) or `Predictions/J` (Track 2) (D5, D9).

2.1 Single-Stream Reactivity versus Batched Amortization

In SemiAnalysis `InferenceMAX` [1], a sports car concentrates all engine power on one passenger for rapid responsiveness while a transit bus amortizes fuel over many passengers for high aggregate throughput. Embodied serving spans two matching regimes across $C \in \{1, 2, 4, 8, 16\}$:

1. **Single-Stream Reactive Regime ($C = 1$).** Dedicating one chip to a single robot stream maximizes $X(1) = 1000/L_{p99}(1)$ and minimizes closed-loop actuation delay $d = \lceil L/\Delta t \rceil$ for rapid slip recovery [10, 17]. At $C = 1$, 2B to 3.3B weights leave execution heavily memory-bandwidth-bound.

2. **Batched Multi-Stream and Parallel Planning Regime ($C \in \{2, 4, 8, 16\}$).** Higher C arises when serving multi-arm robot cells, sampling parallel action chunks as in `BID` [10], or scoring C candidate action trajectories in Model Predictive Control with `Cosmos-Predict2.5-2B` [25]. Weight reuse raises `Actions/s`, `Predictions/s`, `Actions/J`, and `Predictions/J` while shifting $X(C)$ leftward until batch latency breaches the track's gapless $p99$ SLA.

2.2 Discrete Concurrency Sweeps and Non-Dominated Frontiers

Single-point metrics encourage batch tuning that degrades control latency. Under **D4**, entrants evaluate one engine per track across the ordered canonical grid $C \in \{1, 2, 4, 8, 16\}$ starting at $C = 1$ with untimed keep-alive drain and the $\leq 15\%$ per-slot completion fairness fence. Retaining SLA-compliant discrete points yields an audited latency-throughput frontier with zero interpolation between unmeasured concurrencies.

3. Differences Between Token Streaming and Action Chunking

In **SemiAnalysis InferenceX** [1, 2] and **AgentX** [3], autoregressive LLM decoding emits one token per forward pass per sequence (or $1 + \gamma$ accepted tokens under speculative decoding). Embodied control departs from token streaming along distinct horizon and delay axes.

3.1 Prediction Horizon versus Locked Execution Horizon

Visuomotor policy architectures originating with **ACT** [4] and **Diffusion Policy** [5] separate two temporal horizons:

- **Prediction Horizon (H or T_p).** Total future actions predicted by the network head in one forward pass ($H = 100$ at 50 Hz in **ACT** [4]; $H = 16$ in **Diffusion Policy** [5]; $H = 10$ on LIBERO in π_0 [6] and $\pi_{0.5}$ [7]; $H = 16$ in **GR00T N1 / N1.7-3B** [8]; $H = 8$ in **OpenVLA-OFT** [9]; DCT horizons in **FAST** [46]). Longer prediction windows smooth multimodal demonstrations during training.
- **Execution Horizon (K_{eval} or T_a, $1 \leq K_{\text{eval}} \leq H$).** Number of actions sent to the robot controller before capturing a fresh observation and launching the next inference. The remaining $H - K_{\text{eval}}$ predictions are discarded or blended [4, 10]. EmbodiedX locks K_{eval} across both `Closed` and `Open` divisions ($K_{\text{eval}} = 5$ on Track 1A, $K_{\text{eval}} = 8$ on Track 1B, $K_{\text{eval}} = 16$ frames per rollout prediction on Track 2).

Why Fleet Throughput Counts K_{eval} Instead of H. Defining throughput as raw emitted tensor length $H/\bar{L}$ would grant a DROID $\pi_{0.5}$ head ($H = 50, K_{\text{eval}} = 5$) [7] a 5× higher score than the LIBERO $\pi_{0.5}$ head ($H = 10, K_{\text{eval}} = 5$) and 3.125× higher than `GR00T N1.7-3B` ($H = 16, K_{\text{eval}} = 8$) [8] at identical forward latency simply for outputting unused tail elements. Counting K_{eval} measures physical actions applied to the robot.

3.2 Actuation Delay and Physical Observation Staleness

Standard manipulation suites (**LIBERO** [11], **SimplerEnv** [12], **CALVIN** [13], **BEHAVIOR-1K** [14]) freeze physics while the policy runs, and distributed testbeds such as **RoboArena** [47] restrict tasks to quasi-static regimes where remote RPC latency can be ignored. Real-time robot learning research establishes why zero-delay simulation masks control failure:

- **Delayed-MDP Theory. Ramstedt & Pal** [15], **Xiao et al.** [16], **DCAC** [49], and **Riemer et al.** [50] formulate concurrent control as a delayed Markov Decision Process whose environment evolves throughout inference.
- **Real-Time Action Chunking. RTC (Black, Galliker, & Levine, NeurIPS 2025)** [10] proves that latency L at servo period Δt delays chunk arrival by $d = \lceil L/\Delta t \rceil$ steps; naive switching or **Temporal Ensembling** [4] triggers protective torque stops on bimanual arms under multi-step delay. **Streaming Flow Policy** [51] emits actions progressively along ODE trajectories to lower first-action delay.
- **Delay-Sweep Collapse. VLASH** [17], **TT-RTC** [18], and **Agouzoul (2026)** [19] sweep $d \in [0, 20]$ ticks on LIBERO and Kinetix and document steep drops in task success once observation staleness exceeds contact transition windows.
- **Wall-Clock Evaluation Harnesses. AllenAI `vla-evaluation-harness`** [20] steps simulators against wall-clock time to quantify real-time degradation and stale-action fractions.

4. Prior Literature and Benchmark Landscape

We synthesize 68 papers across six research areas supporting the EmbodiedX v1.0.0 specification.

4.1 Action-Chunked VLA Policies and Unit Disambiguation

Generalist robot policies divide into three model families, all built around action chunking ($H > 1$):

1. **CVAE and Parallel Regression Policies. ACT** [4] predicts $H = 100$ joint positions via a CVAE transformer. **OpenVLA-OFT** [9] ($H = 8$, 72.9 ms on A100), 1-bit **BitVLA** [56] ($K = 25$, 73 ms), and dual fast-in-slow **FiS-VLA** [57] ($K = 8$, 68.0 ms on RTX 4090) emit parallel action chunks in one pass.
2. **Diffusion and Flow-Matching Policies. Diffusion Policy** [5] integrates $T_p = 16$ denoising steps while executing $T_a = 8$. Track 1A model $\pi_{0.5}$ [7] (extending π_0 [6], evaluated at `lerobot/pi05_libero_finetuned_v044`) pairs PaliGemma with a 300M flow-matching expert integrating 10 Euler ODE steps ($H = 10$, locked $K_{\text{eval}} = 5$ at dual 224×224 RGB on `HuggingFaceVLA/libero`). Track 1B model `GR00T N1.7-3B` [8] (evaluated at `nvidia/GR00T-N1.7-LIBERO`, `libero_10`)

couples a `Cosmos-Reason2-2B` VLM with a 16-step DiT flow action head ($H = 16$, locked $K_{\text{eval}} = 8$ at dual 256×256 RGB on `HuggingFaceVLA/libero`). **TinyVLA** [58] and **SmolVLA** [24] target compact GPUs, and **Streaming Flow Policy** [51] streams actions mid-integration.

3. **Token Compression and Runtime Acceleration.** **FAST** [46] applies DCT+BPE token compression. Methods surveyed by **Guan et al.** [52] and **Yu et al.** [53] speed up VLA forward passes via token reuse (**VLA-Cache** [42]), early exit (**DeeR-VLA** [33]), token and layer pruning (**EfficientVLA** [43]), action reuse (**FlashVLA** [44]), and speculative decoding (**Spec-VLA** [45]), while **Zhang et al.** [54] measure divergence between physical robot completion time and frozen-simulator success rates.

Disambiguating the Three Uses of "Hz". Across 2024–2026 VLA papers, authors attach "Hz" to three incompatible quantities separated by K ($5\times$–$25\times$): **Camp A ("Hz"** $= K/L$**)** reports open-loop action emission (**OpenVLA-OFT** [9]: 72.9 ms, K=8 → 109.7 Hz; **BitVLA** [56]: 321 ms, K=25 → "77.9 Hz" and 73 ms → 341.1 Hz; **FiS-VLA** [57]: 68 ms, K=8 → 117.7 Hz); **Camp B ("Hz"** $= 1/L$**)** reports chunk rate (**VLA-Cache** [42]: 14.66 Hz; **EfficientVLA** [43]: 19.96 Hz); and **Camp C** separates forward latency in ms from the fixed $1/\Delta t = 20$–50 Hz servo clock ($\pi_0/\pi_{0.5}$/**RTC**/**TT-RTC** [6, 7, 10, 18], **GR00T** [8], **TinyVLA** [58], **SmolVLA** [24], **VLASH** [17], **A2C2** [55]). EmbodiedX plots **`Chunks/s`** ($1000/L_{p99}$) on X and **`Actions/s`** on Y: `OpenVLA-OFT` at 72.9 ms (K=8) records $X = 13.7$ Chunks/s and $Y(1) = 109.7$ Actions/s, so a 321 ms model at K=25 ($X = 3.1$ Chunks/s) never appears more reactive than a 50 ms policy ($X = 20.0$ Chunks/s).

4.2 Asynchronous and Real-Time VLA Execution

In **RTC** [10], test-time flow inpainting (`IT-RTC`) smooths delayed chunk boundaries through vector-Jacobian product (`vjp`) backpropagation across the flow expert (+27% latency overhead). **TT-RTC** [18] and closed-form blending eliminate test-time `vjp` overhead, motivating **Decision D6** (forward-only asynchronous overlap in `LIBERO-RT` compatible with compiled `TorchTPU`, `XLA`, and `TensorRT` engines). **VLASH** [17] rolls proprioceptive state forward to $t+d\Delta t$, and **A2C2** [55] applies per-step residual corrections. Where prior control studies inject synthetic step delays d on one stream and **AllenAI** **`vla-evaluation-harness`** [20] uses host OS sleeps without concurrency sweeps or power sensors, EmbodiedX ties measured hardware tail latency across $C \in \{1, 2, 4, 8, 16\}$ directly to gapless SLAs and latency-coupled simulation.

4.3 Age of Information in Feedback Control

Kaul, Yates, & Gruteser (IEEE INFOCOM 2012) [21] define instantaneous Age of Information $\Delta(t) = t - u(t)$ from sensor capture timestamp $u(t)$ and prove that minimizing $\Delta(t)$ differs mathematically from maximizing throughput or minimizing queueing delay alone. **Costa, Codreanu, & Ephremides (IEEE TIT 2016)** [22] introduce **Peak Age of Information (PAoI)** to bound worst-case control excursion immediately prior to update arrival, and **Yates & Kaul (IEEE TIT 2019)** [48] plus **Yates et al. (IEEE JSAC 2021)** [23] extend PAoI to C streams sharing one server in closed-loop estimation and LQG control.

4.4 Manipulation and Perturbation Robustness Benchmarks

Static manipulation suites (**LIBERO** [11], **SimplerEnv** [12], **CALVIN** [13], **BEHAVIOR-1K** [14]) omit dynamic scene perturbations. **LIBERO-Plus (Fei et al., 2025)** [28] documents VLA success falling from 95% below 30% under camera and pose shifts, and **VLA-Arena (ICML 2026)** [29] introduces graded visual and language perturbations across 170 tasks. EmbodiedX integrates input sensitivity directly into timed runs via cryptographic pixel micro-jitter and blind inline G4 perturbation probes on real `HuggingFaceVLA/libero` and `lerobot/droid_100` frames (D10).

4.5 World-Model Regimes and Video Simulation Benchmarks

Interactive world models fall into four regimes: **(1) Causal Streaming Video DiTs** (**Diffusion Forcing** [59], **CausVid** [60], **Self Forcing** [61]) built with causal student distillation and rolling KV caches; **(2) Frame-Causal Game Engines** (**GameNGen** [62], **Matrix-Game 2.0** [63]) at 20–25 FPS; **(3) Bidirectional Full-Clip 3D-DiTs** (`Cosmos-Predict2.5-2B` [25], **Sparse VideoGen** [64], **Radial Attention** [65]), where all $K_{\text{eval}} = 16$ latent frames attend bidirectionally before 3D-VAE decoding so TTFF $\equiv L_{\text{clip}}$; and **(4) Batched MPC Planners** (**DINO-WM** [66], **Genie Envisioner** [27]) evaluating C action futures in parallel. Existing benchmarks (**1X Challenge** [36], **EWMBench** [38], **WorldScore** [39], **WorldArena 1.0/2.0** [40, 41]) score offline fidelity without real-time SLAs, batch sweeps, or energy telemetry. EmbodiedX evaluates `nvidia/Cosmos-Predict2.5-2B` (`robot/action-cond`) strictly in `Predictions/s` and `Predictions/J` ($K_{\text{eval}} = 16$ frames per rollout prediction) under a gapless deadline of $L_{p99,\text{WM}} \leq 1000.0$ ms (≥ 1.0 Pred/s), and permits step or causal distillation in `Open` when backed by inline parity and Stage-2 visual fidelity and dynamics ranking checks.

4.6 ML Systems Energy and Roofline Methodology

MLPerf Inference [35] defined `Closed`/`Open` divisions and `LoadGen` isolation, while **SemiAnalysis InferenceX v1/v2** [1, 2] and **AgentX** [3] standardized concurrency-swept Pareto curves. In robotics systems, **MAVBench** [67] studied UAV latency-energy coupling, **RobotPerf** [37] benchmarked ROS 2 operator graphs, and **ManiSkill3** [68] scaled GPU-parallel simulator throughput without policy tail-latency sweeps. Energy and Roofline normalizations follow **MLPerf Power** [30], **ML.ENERGY** [31], **AI Energy Score** [32], **DeeR-VLA** [33], and **Williams et al.** [34].

4.7 Cross-Benchmark Comparison Matrix

Benchmark / Framework	Domain	Latency-Coupled Sim	Locked K_{eval} + PAoI	Discrete C Pareto	Metered Energy	Anti-Gaming + Quality	Std. Cloud GPU + TPU
LIBERO [11] / **SimplerEnv** [12] / **CALVIN** [13]	VLA Manipulation	No (Frozen)	No	No	No	Yes (Static SR)	No
ManiSkill3 [68] / **BEHAVIOR-1K 2026** [14]	GPU Sim / Household VLA	No (Sim time)	No	Partial (Sim FPS)	No	Yes (SR/Q-score)	No (1 GPU)
MAVBench [67] / **RobotPerf** [37]	UAV / ROS 2 Compute	Partial (Pre-VLA)	No	No	Partial (MAVBench)	No (No VLA/WM)	Partial (CPU/GPU/FPGA)
LIBERO-Plus [28] / **VLA-Arena** [29]	VLA Perturbations	No (Frozen)	No	No	No	Yes (Perturb SR)	No
RTC [10] / **TT-RTC** [18] / **VLASH** [17] / **A2C2** [55]	Real-Time VLA	Yes (Step delay d)	Partial (Synthetic d)	No ($C = 1$/BID)	No	Yes (Delayed SR)	No (1 GPU)
AllenAI `vla-eval-harness` [20]	Real-Time VLA	Yes (Wall sleep)	Partial (Stale ratio)	No	No	Partial (Off-table)	No (1x H100)
EWMBench [38] / **WorldScore** [39] / **WorldArena** [40, 41]	Embodied World Models	No (Offline)	No	No	No	Yes (Visual/Dyn)	No
MLPerf Inference & Power [30, 35]	General ML Systems	No (Open-loop)	No	Partial (2 modes)	Yes (PTDaemon)	No (Separate run)	Yes (Multi-vendor)
SemiAnalysis InferenceX / AgentX [1, 2, 3]	LLM / Agentic Serving	No (Token stream)	No	Yes (Full C sweep)	Partial (Modeled)	No (Separate evals)	Yes (GPU/TPU)
ML.ENERGY [31]	GenAI Energy	No (Open-loop)	No	Yes (Batch sweep)	Yes (GPU `J/req`)	No	Partial (GPUs)
EmbodiedX v1.0.0 (`embodied-efficiency-bench`)	**Real-Time VLA & World Models**	Yes ($d = \lceil L/\Delta t \rceil$)	Yes ($K_{\text{eval}} \in \{5, 8, 16\}$)	Yes ($C \in \{1..16\}$)	Yes (`Act/J`, `Pred/J`)	Yes (**Inline + Stage-2 + C-R**)	Yes (**L40S & TPU v6e-1**)

Table 1: Comparison of EmbodiedX v1.0.0 against prior robotics, world-model, and ML inference benchmarks.

5. Axis Formulation and Hardware Normalizations

5.1 Guaranteed Interactivity and Peak Age of Information

5.1.1 Pooled Steady-State Tail Rate and Fairness Fence

To follow **SemiAnalysis InferenceX** [1, 2] (**Right on X = Faster**) while keeping the 99th percentile statistically well-conditioned at high concurrency ($N \geq 320$ chunks on Track 1, $N \geq 200$ rollout predictions on Track 2) and blocking LIFO slot starvation, EmbodiedX defines X over pooled steady-state $L_{p99}(C)$ across active slots paired with a mandatory $\leq 15\%$ per-slot completion fairness fence (D4, D5, D10):

$$X_{\text{VLA}}(C) = \frac{1000}{L_{p99}(C)} \ \text{[Chunks/s]}, \qquad X_{\text{WM}}(C) = \frac{1000}{L_{p99,\text{WM}}(C)} \ \text{[Predictions/s]} \tag{3}$$

At $\Delta t = 50$ ms (20 Hz), the gapless real-time cutoffs lie at $X_{\text{SLA}} = 4.0$ Chunks/s ($L_{p99} \leq 250.0$ ms, Track 1A $K_{\text{eval}} = 5$), $X_{\text{SLA}} = 2.5$ Chunks/s ($L_{p99} \leq 400.0$ ms, Track 1B $K_{\text{eval}} = 8$), and $X_{\text{SLA}} = 1.0$ Pred/s ($L_{p99,\text{WM}} \leq 1000.0$ ms, Track 2 $K_{\text{eval}} = 16$ frames per rollout prediction), while 20.0 Chunks/s (50.0 ms) marks single-tick reaction.

5.1.2 Peak Age of Information Derivation

For sensor capture time $u(t)$, instantaneous observation age at the actuator is $\Delta(t) = t - u(t)$ [21]. Under **D7**, the harness computes worst-case **Peak Age of Information (PAoI)** [22, 23] on the final executed step ($k = K_{\text{eval}}$) across steady-state chunks m:

$$\Delta_{\text{peak}}(C, K_{\text{eval}}) = \text{p99}_m\big(t_{\text{act}}(m, K_{\text{eval}}) - t_{\text{obs}}(m)\big) = L_{p99}(C) + (K_{\text{eval}} - 1)\Delta t_{\text{eff}} \quad \text{[ms]} \tag{4}$$

Whenever an engine meets its gapless deadline ($L_{p99}(C) \leq K_{\text{eval}}\Delta t$) under forward-only asynchronous overlap (D6), zero inter-chunk queue stall occurs ($\Delta t_{\text{eff}} = \Delta t = 50$ ms). Under locked K_{eval}, $\Delta_{\text{peak}}(C, K_{\text{eval}})$ shifts $L_{p99}(C)$ by a constant offset, so ranking by $X_{\text{VLA}}(C)$ preserves exact PAoI ordering within each track.

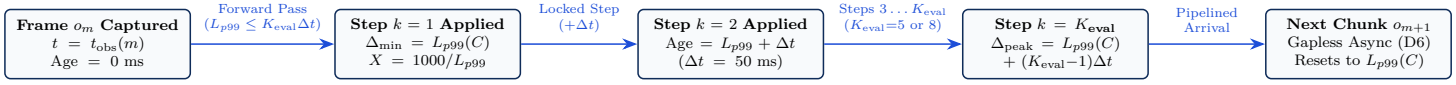


Figure 1: Guaranteed interactivity $X(C) = 1000/L_{p99}(C)$ (D5) and co-reported Peak Age of Information $\Delta_{\text{peak}}(C, K_{\text{eval}})$ (D7) under locked horizon $K_{\text{eval}} \in \{5, 8\}$ and gapless asynchronous overlap (D6).

5.2 Fleet Throughput and Hardware Normalizations

5.2.1 Steady-State Action and Prediction Throughput

Across canonical grid $C \in \{1, 2, 4, 8, 16\}$ over the steady-state full-occupancy window $T_{\text{steady}}(C)$ (kept saturated via untimed keep-alive requests and verified by the $\leq 15\%$ per-slot completion fairness fence), primary Fleet Throughput is:

$$Y_{\text{VLA}}(C) = \frac{N_{\text{timed}}(C) \cdot K_{\text{eval}}}{T_{\text{steady}}(C)} \ \text{[Actions/s]}, \qquad Y_{\text{WM}}(C) = \frac{N_{\text{timed}}(C)}{T_{\text{steady}}(C)} \ \text{[Predictions/s]} \tag{5}$$

where each Track 2 rollout prediction generates $K_{\text{eval}} = 16$ future frames, and 20 Actions/s corresponds to one gapless 20 Hz VLA control stream.

5.2.2 Power and Roofline Normalization Views

From the identical run receipt, EmbodiedX computes the two plotted Season 1 leaderboard views (`Throughput vs. Interactivity` and `Efficiency vs. Interactivity`) plus two auxiliary schema normalizations (D5):

Y-Axis View	Physical Formula	Literature Basis	Role in EmbodiedX Season 1
1. Fleet Throughput *(`Actions/s`, `Predictions/s`)*	$\frac{Y_{\text{VLA}}(C) \text{ or } Y_{\text{WM}}(C)}{1 \text{ chip}}$	InferenceX [1] MLPerf [35] RobotPerf [37]	Official **Real-Time Fleet Throughput** award metric over SLA-compliant discrete points in $\{1, 2, 4, 8, 16\}$ (D9).
2. Green Efficiency *(`Actions/J`, `Predictions/J`)*	$\frac{Y(C)}{\bar{P}_{\text{board}}\ (\text{W})} = \frac{\text{Output}}{\text{Joule}}$	MLPerf Power [30] ML.ENERGY [31] AI Energy Score [32] DeeR-VLA [33]	Official **Green Embodied Efficiency** award metric requiring `power_provenance == "MEASURED"` via constant-cap `10 Hz NVML` board power on `NVIDIA L40S` (`--assume-power-watts` rejected; `TPU v6e-1 = N/A`).
3. Ref. Hourly Rate *(Auxiliary `(Out/s)/($/hr)`)*	$\frac{Y(C)}{\text{Ref. Rate (\$/hr)}}$	InferenceX v2 [2]	Auxiliary schema diagnostic against pinned on-demand cloud rates (`~$0.88-$1.15/hr L40S`, `~$1.375/hr TPU v6e-1`).
4. Roofline Iso-Views *(Auxiliary `/100 TF`, `/TB/s`)*	$\frac{Y(C)}{P_{\text{BF16}}/100}, \frac{Y(C)}{\text{BW}_{\text{peak}}}$	Williams et al. [34]	Auxiliary Roofline efficiency metric against dense BF16 peak and memory BW (362.05 TF / 864 GB/s `L40S`; 918 TF / 1,600 GB/s `TPU v6e-1`).

Table 2: Primary plotted Season 1 leaderboard metrics (Views 1–2) and auxiliary Roofline/cost diagnostics (Views 3–4).

6. Hardware Targets Verification Protocol and Season 1 Governance

6.1 Absolute Physical Units Over Relative Speedups

Scoring relative speedup ($L_{\text{base}}/L_{\text{sub}}$) penalizes platforms with well-tuned reference compilers and prevents cross-platform comparison on a shared physical scale [1, 35]. EmbodiedX ranks submissions exclusively in absolute SI units (`Chunks/s`, `Predictions/s`, `Actions/s`, `Actions/J`, `Predictions/J`) with zero organizer-seeded vendor numbers.

6.2 Standardized Season 1 Cloud Hardware Targets

Every canonical workload (2B to 3.3B parameters) fits comfortably on one device. Under **D8**, Season 1 standardizes strictly on two single-chip (`num_devices == 1`, TP = PP = EP = 1) cloud accelerators at stock vendor TDP and clock limits so entries compete on device kernels, graph compilers, low-bit quantization, and request scheduling while organizers can re-run finalists on identical silicon:

1. **Standard GPU (`nvidia_l40s_48gb`). `NVIDIA L40S`** (Ada Lovelace SM89, 48 GB GDDR6 at 864 GB/s, 350 W board TDP, native FP8 Tensor Cores, 362.05 TF dense BF16, active `10 Hz NVML` board power telemetry). Available via Shadeform (`~$0.88-$1.15/hr`), AWS EC2 `g6e.xlarge`, or the NVIDIA Academic Grant Program.
2. **Standard TPU (`tpu_v6e_1_32gb`). `Google Cloud TPU v6e-1`** (Trillium single-chip slice, 32 GB HBM2e at 1,600 GB/s, 918 TF dense BF16, 1,836 TOPS INT8/FP8; isolate multi-chip hosts via `TPU_CHIPS_PER_HOST_BOUNDS=1,1,1`, `TPU_HOST_BOUNDS=1,1,1`, and `TPU_VISIBLE_DEVICES=0`). Available via Google Cloud (`~$1.375/hr`) or free through a renewable 30-day Google TPU Research Cloud (TRC) allocation.

Additional single-chip and edge profiles remain in the hardware schema solely for local development and future onboard-edge seasons.

6.3 Closed and Open Quality Gates and Scalar Award Categories

Because crossing Pareto curves prevent a single total order, Season 1 crowns winners across three scalar categories inside each independent track pool (`Track 1A`, `Track 1B`, `Track 2`):

Season 1 Award Category	Scalar Ranking Metric	Mandatory Admissibility Gate
1. Interactive Reactivity (`C = 1`)	Highest single-stream rate $1000/L_{p99}(1)$ at $C = 1$ (lowest C=1 $p99$ latency, tie-broken by lowest $p50$ latency).	Unimodal tail $L_{p99}(1)/L_{p50}(1) \leq 1.5$ plus full division quality gates (≥ 100 golden samples).
2. Real-Time Fleet Throughput	Highest aggregate `Actions/s` (Track 1) or `Predictions/s` (Track 2) across canonical points $C \in \{1, 2, 4, 8, 16\}$.	Selected point must meet the gapless pooled $p99$ SLA (≤ 250, ≤ 400, or ≤ 1000 ms) and $\leq 15\%$ slot fairness fence.
3. Green Embodied Efficiency	Highest hardware-measured `Actions/J` (Track 1) or `Predictions/J` (Track 2) over SLA-compliant grid points.	`power_provenance == "MEASURED"` via constant-cap `10 Hz NVML` board power on `L40S` (`--assume-power-watts` rejected; `TPU v6e-1 = N/A`).

Table 3: Official EmbodiedX Season 1 scalar award categories per track. Exact scalar ties are broken by lower-step Real-Time Dominated Hypervolume wrt $(0.5X_{\text{SLA}}, 0)$ over SLA-compliant discrete points in $\{1, 2, 4, 8, 16\}$ without curve interpolation.

6.4 Anti-Gaming Defenses and Two-Phase Commit-Reveal

1. **Eight-Gate Anti-Gaming Verifier (D4, D10).** Every bundle must pass eight automated checks: (i) single-chip isolation (`num_devices == 1`) at stock TDP/clocks; (ii) canonical ordered grid $C \in \{1, 2, 4, 8, 16\}$ from $C = 1$ with monotonic OOM truncation only; (iii) locked K_{eval} and real public traces from `HuggingFaceVLA/libero` and `lerobot/droid_100` at native wire payloads ($\geq 224 \times 224$ RGB, zero synthetic sine waves); (iv) opaque random `UUIDv4` `session_id`s plus dynamically seeded per-frame cryptographic $\pm 1/255$ `uint8` pixel micro-jitter banning hash lookup tables; (v) outlier fence max $/p50 \leq 4.0$, per-slot completion fairness skew $\leq 15\%$, and C=1 tail unimodality $p99/p50 \leq 1.5$; (vi) $N = 3$ trial consistency ($\leq 8\%$ pooled $p99$ spread across $N \geq 320$ VLA / $N \geq 200$ WM calls after `--warmup >= 30` on the live model loop; lock SM clocks via `nvidia-smi -lgc` on the real workload only if boost trips variance); (vii) inline Golden-IO parity over `--num-samples >= 100` real samples (CosSim ≥ 0.995 + CVaR-1% on Track 1; SSIM ≥ 0.88, PSNR ≥ 26.0 dB, CVaR-1%(SSIM) ≥ 0.80, CVaR-1%(PSNR) ≥ 22.0 dB on Track 2) and G4 sensitivity probes (`Closed`) or signed Stage-2 quality receipt (`Open`); and (viii) SHA-256 receipt integrity and constant-cap `MEASURED` NVML power verification (`torch.cuda.synchronize()`, `torch.tpu.synchronize()`, or `jax.block_until_ready()` before stopping request timers).
2. **Two-Phase Commit-Reveal and Clean-Hardware Re-run.** In Phase 1, contestants package and verify public pull request bundles via `eeb prepare-commit-bundle` and `eeb verify-bundle --strict`, submitting only `bundle_summary.json` and `submission_meta.json` carrying `recipe_commit_sha256`, the SHA-256 digest over canonical `reveal_manifest.json` (pinning immutable OCI container image `@sha256`, Hugging Face checkpoint commit SHA and shard SHA-256 digests, lockfile SHA-256, and CLI command). Within 24 hours after December 18, 2026 (`23:59 AoE`), shortlisted finalists (at most one finalist slot per lab or organization per track and division) privately supply their matching container and manifest. Final Official Season 1 Awards are ranked strictly by Phase-2 organizer clean-hardware re-runs on a private held-out split; agreement within $\pm 8\%$ of Phase 1 serves strictly as an anti-fraud disqualification gate. Confirmed Season 1 winners publish open recipes (`Verified Open Recipe`); rolling `2027+` entries support both open and `Proprietary / Audit-Only` attribution.

6.5 End-to-End Pipeline Validation Matrix and Season 1 Schedule

All reference servers, public trace loaders, telemetry sidecars, and eight-gate validators were verified end-to-end on physical `NVIDIA L40S` and `Google Cloud TPU v6e-1` prior to release. Under Option A, Season 1 opens with 0 seeded entries across all tracks (Table 4).

Track, Architecture, & Checkpoint	Horizon, Traces, & SLA	Division Quality Gates (≥ 100 Samples)	Validated Standard GPU (L40S) & TPU (v6e-1) Stacks	Seeded
Track 1A: $\pi_{0.5}$**-LIBERO** `lerobot/pi05_libero_finetuned_v044` `@7152f13e34c6`	H=10, **locked** K_{eval}=5 `HuggingFaceVLA/libero` Dual 224^2 RGB, 10 steps $L_{p99} \leq 250.0$ ms (≥ 4.0 Chunks/s)	**Closed:** CosSim ≥ 0.995 + CVaR-1% **Open:** Stage-2 `LIBERO-RT` within -5.0 pp	**L40S:** `PyTorch (eager/compile) + 10 Hz NVML` **v6e-1:** `openpi JAX (jit)` & `TorchTPU`; 8 gates verified	0
Track 1B: `GR00T N1.7-3B` `nvidia/GR00T-N1.7-LIBERO` (`libero_10`, `@2ea293aa20ba`)	H=16, **locked** K_{eval}=8 `HuggingFaceVLA/libero` Dual 256^2 RGB, 16 steps $L_{p99} \leq 400.0$ ms (≥ 2.5 Chunks/s)	**Closed:** CosSim ≥ 0.995 + CVaR-1% **Open:** Stage-2 `LIBERO-RT` within -5.0 pp	**L40S:** `PyTorch (eager/compile) + 10 Hz NVML` **v6e-1:** `TorchTPU (eager/compile)`; 8 gates verified	0
Track 2: `Cosmos-Predict2.5-2B` `nvidia/Cosmos-Predict2.5-2B` (`robot/action-cond`, `@85f8ae7bfe8f`)	**Locked** K_{eval}=16 frames `lerobot/droid_100` 256^2 RGB, 10 steps, CFG=1 $L_{p99} \leq 1000$ ms (≥ 1.0 Pred/s)	**Closed:** SSIM ≥ 0.88, PSNR ≥ 26.0 dB, CVaR-1% $\geq 0.80/22.0$ dB **Open:** + LPIPS ≤ 0.12, FVD ≤ 120, InvDyn ≥ 0.99, $\tau \geq 0.80$	**L40S:** `PyTorch (eager/compile) + 10 Hz NVML` **v6e-1:** `TorchTPU` & `JAX` servers; 8 gates verified	0

Table 4: End-to-end validated pipeline and target platform specification matrix for EmbodiedX Season 1. Full 40-character Hugging Face commit SHAs appear in Section 1.1 (D2). Season 1 opens with 0 pre-seeded vendor entries so community pull requests establish the founding Pareto frontier.

Season 1 Calendar and Winner Recognition. Season 1 launches **October 30, 2026**. **Checkpoint 1 (November 10, 2026, 23:59 UTC)** freezes entries featured live on stage as Founding Frontier Leaders at the CoRL 2026 EEAI Workshop in Austin on **November 12, 2026**. **Checkpoint 2 (December 18, 2026, 23:59 AoE)** closes Season 1 awards (+24h finalist reveal; continuous rolling leaderboard in `2027+`). Verified Season 1 category winners receive **monetary prizes**, an official **CoRL 2026 EEAI Workshop Award Certificate**, **an invitation to apply for Google Cloud credits**, permanent verified leaderboard attribution, and an invitation to co-author the post-season *EmbodiedX Season 1 Systems Report*.

6.6 Planned Extensions Beyond Season 1

Future seasons introduce planned additions (S2-1 and S2-2): (1) variable-horizon planning evaluated directly against Peak Age of Information $\Delta_{\text{peak}}(C, K)$ alongside sub-100 ms frame-causal streaming world-model gates [59, 60, 61, 62, 63], and (2)

dedicated onboard edge SoC divisions, multimodal omni policies (`Qwen3-Omni-30B-A3B` [26]), and physical robot evaluation arenas (`RoboArena` [47]).

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